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IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

GAME THEORY FOR MULTI-AGENT DECISION MAKING

Monday, 19 May 2025, 10:00 AM

Time allowed: 3:00 hours

There are FIVE questions on this paper.

Answer FOUR questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : R. Vinter
Second Marker(s) : Y. Gao

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MSc and EEE Part IV: M.Eng. and ACGI

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Information for candidates:

No special instructions for candidates.

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1(A). Consider the 2-player matrix game in mixed strategies, with $n \times m$ pay-off matrices A and B. Specifically, when players 1 and 2 choose mixed strategies $x \in \Lambda^{(n)}$ and $y \in \Lambda^{(m)}$, their pay-offs are

$$x^T A y \text{ and } x^T B y$$

respectively. (Both players aim to minimize their pay-offs.) Here $\Lambda^{(n)}$ is the set of discrete probability weights

$$\Lambda^{(n)} := \{x = [x_1, \ldots, x_n]^T \ : \ x_i \geq 0 \text{ for } i = 1, \ldots, n \text{ and } \sum_{i=1}^{n} x_i = 1\}.$$

Let $(\bar{x}, \bar{y})$ be a Nash equilibrium for this game. Write

$$\bar{p} := -\bar{x}^T A \bar{y} \text{ and } \bar{q} := -\bar{x}^T B \bar{x}\,.$$

Show that $(\bar{x}, \bar{y}, \bar{p}, \bar{q})$ solves the nonlinear program

$$(NL) \begin{cases} \text{Minimize } J(x, y, p, q) := x^T (A + B) y + p + q \\ \text{over } x \in \Lambda^{(n)}, y \in \Lambda^{(m)} \text{ and numbers } p \text{ and } q \text{ satisfying} \\ Ay \geq -p 1_n \text{and } x^T B \geq -q 1_m^T. \end{cases}$$

Here $1_n^T := [1, \ldots, 1]$.

Hint: Show, as a first step, that, for any elements (x, y, p, q) satisfying the constraints of (NL), we have $J(x, y, p, q) \geq 0$. [6]

Give conditions on the matrices A and B for (NL) to be a convex optimization problem and comment very briefly on the significance that (NL) has this property. [2]

1(B). Now consider the matrix game in mixed strategies of part (A), in which the pay-off matrix for the player 1 is

$$A = \begin{bmatrix} 0 & 1 & 2 & -2 \\ 1 & 0 & -2 & 2 \end{bmatrix}.$$

By using graphical methods, or otherwise, to find a (mixed) safety strategy $(1 - \bar{\lambda}, \bar{\lambda})$ for player 1. [12]

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2. Consider a two-player game, with players A and B. Each player can choose L ('left') or R ('right'). Both players seek to minimize their pay-offs.

The key features of the information available to the players are as follows: A does not know B's chosen strategy. B can choose to pay in order to find out A's chosen strategy. Specifically

- B knows A's chosen strategy if B pays ($a > 0$ dollars)
- B does not know A's chosen strategy if B does not pay.

The pay-offs and information patterns for the two scenarios ('B pays' and 'B does not pay') are as indicated in Fig. 2a and Fig. 2b, respectively.

Show that the game can be set up as a normal form matrix game for players A and B. [7]

Hint: the policy space for B contains 6 elements, to take account of B's choice to pay, or not, and, if B pays, the different ways B can react to knowledge of A's chosen strategies.

Determine the entries of the pay-off matrices of the normal form game for the two players, [7]

Identify any Nash equilibria, and state whether they are admissible. [4]

Comment on whether, in this game, it is advantageous to B to buy access to information about A's strategy. [2]

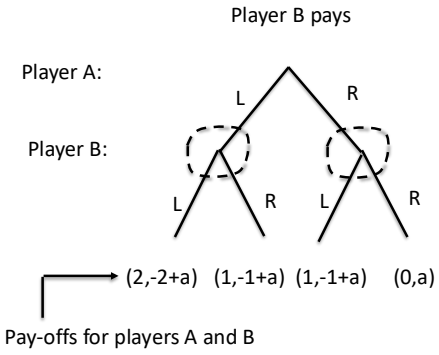


Fig. 2(a)

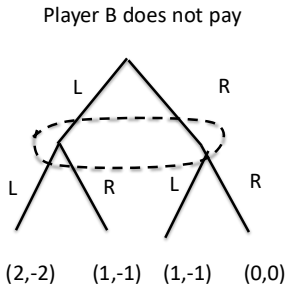


Fig. 2(b)

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3. *(Asymmetric Cournot Triopoly)* Three companies, A , B, and C make the same product and charge the same price. However their production costs are different. Assume

$$\begin{array}{ccc} \text{Company A} & \text{Company B} & \text{Company C} \\ \begin{cases} \text{level of production} = a \\ \text{cost of production} = 2a \end{cases} & \begin{cases} \text{level of production} = b \\ \text{cost of production} = 4b \end{cases} & \begin{cases} \text{level of production} = c \\ \text{cost of production} = 4c \end{cases} \end{array} .$$

where $a > 0$, $b > 0$, and $c > 0$. The price is related to the levels of production a, b and c, according to formula

$$\text{Price} = 12 - a - b - c\,.$$

The profit of each company, which it seeks to maximize, is as given by:

$$\text{price} \;\times\; \text{level of production} \;-\; \text{production costs.}$$

Assume that each of the three companies independently chooses its production level, i.e., any company does not know the production level of the other companyies before setting its own production level.

(i): Determine the profit functions $L^A(a,b,c)$, $L^B(a,b,c)$, $L^C(a,b,c)$ of the three companies. [2]

(ii): Determine the (unique) collection of production levels $(\bar{a},\bar{b},\bar{c})$ that is a Nash equilibrium. [6]

(iii): Now regard company A as the leader in a Stackelberg game and companies B and C as followers. Determine the optimizing Stackelberg production level a^* for company A and the corresponding production levels b^* and c^* for companies B and C. [8]

(Hint: To simplify the calculations in (ii) and (iii), you may assume that, due to symmetry, $\bar{b} = \bar{c}$ and $b^* = c^*$.)

(iv): The response surfaces for Companies A, B and C are the sets in 3-dimensional space

$$\{(a,b,c) : a = a(b,c)\},\ \{(a,b,c) : b = b(a,c)\} \text{ and } \{(a,b,c) : c = c(a,b)\},$$

in which $a(b,c)$, $b(a,c)$, $c(a,b)$ are the functions:

$$a(b,c) = \arg\min_{a'>0} L^A(a',b,c),\ b(a,c) = \arg\min_{b'>0} L^A(a,b',c),\ c(a,b) = \arg\min_{c'>0} L^A(a,b,c').$$

(Notice that there is a unique minimizer of $L^A(.,a,b)$ so 'arg min' is well-defined, etc.)
Show:

'$(\bar{a},\bar{b},\bar{c})$ lies in the intersection of the 3 response surfaces' $\implies$ '$(\bar{a},\bar{b},\bar{c})$ is a Nash equilibrium'.

[4]

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4(A). Consider a 2-dimensional evolutionary game with fitness functions

$$a_1(x) \text{ and } a_2(x),\ x \in \Lambda^{(2)},$$

in which $\Lambda^{(2)} := \{(x_1, x_2) \in R^2 : \ x_1 \geq 0, x_2 \geq 0 \text{ and } x_1 + x_2 = 1\}$. (Members of each sub-population seek to maximize their fitness.)

(i): Give the definitions of an (evolutionary) equilibrium and of a stable equilibrium. [2]

(ii): Take a strictly interior Nash equilibrium $\bar{x}$. Show that $\bar{x}$ is a stable equilibrium if, for all sufficiently small $\epsilon > 0$,

$$\begin{cases} (a_1(x) - a_1(\bar{x})) - (a_2(x) - a_2(\bar{x})) < 0 \text{ when } x = (\bar{x}_1 + \epsilon, \bar{x}_2 - \epsilon) \\ \\ \text{and} \\ \\ (a_1(x) - a_1(\bar{x})) - (a_2(x) - a_2(\bar{x})) > 0 \text{ when } x = (\bar{x}_1 - \epsilon, \bar{x}_2 + \epsilon). \end{cases}$$

[6]

(iii): What are the replicator equations for these fitness functions? Briefly describe the expected behaviour of the solution $x(t)$, $t \geq 0$, to the replicator equation, with initial condition $(x_1(0), x_2(0))$ near a strictly interior Nash equilibrium, depending on whether or not the Nash equilibrium is stable. [4]

4(B). There are two possible routes for vehicles to travel on a road network. Let the proportion of cars on the two routes be x_1 and x_2, respectively. Assume that the travel times on the two routes are

$$-a_1(x_1, x_2) = 2x_1 + 2x_2^2 \text{ and } -a_2(x_1, x_2) = 3x_2$$

(the minus sign accounts for the fact that drivers aim to minimize, not maximize, travel times.)

Find a strictly interior Nash equilibrium $\bar{x}$. [4]

By using the results of part (A) (ii), or otherwise, show that this equilibrium is stable. [4]

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5(A). For a given number x_0 and sequence of scalar controls $\{u_0, \ldots, u_{N-1}\}$, define

$$J(x_0, \{u_0, \ldots, u_{N-1}\}) := x_N^2 + \sum_{t=0}^{N-1} u_t^2 \,,$$

in which $\{x_0, x_1, \ldots x_N\}$ is the solution of the state equation

$$x_t = x_{t-1} + u_{t-1}, \text{ for } t = 1, \ldots, N,$$

for the given initial state x_0.

Consider the scalar optimal control problem

$$\begin{cases} \text{Minimize } J(x_0, \{u_i\}) \\ \text{over controls } \{u_0, \ldots, u_{N-1}\}. \end{cases}$$

Find a solution to the Bellman equation, of the form $V_t(x) = p_t x^2$, $t = 0, \ldots, N$ for some positive numbers $p_0, \ldots, p_N$ and show that $p_t > 0$ for all t. [8]

Derive equations for the minimizing control $\{\bar{u}_0, \ldots, \bar{u}_{N-1}\}$. [4]

What is the minimum cost? [4]

5(B). Consider a Stackelberg game, related to the optimal control problem of part (A), but now involving a leader and a follower.

The leader chooses an initial condition x_0 and communicates this information to the follower, who chooses a control strategy $\{u_0, \ldots, u_{N-1}\}$. The pay-offs $L^{\text{lead}}(x_0, \{u_0, \ldots, u_{N-1}\})$ and $L^{\text{follow}}(x_0, \{u_0, \ldots, u_{N-1}\})$, which the players seek to minimize, are

$$\begin{cases} L^{\text{follow}}(x_0, \{u_0, \ldots, u_{N-1}\}) = J(x_0, \{u_0, \ldots, u_{N-1}\}) \\ L^{\text{lead}}(x_0, \{u_0, \ldots, u_{N-1}\}) = J(x_0, \{u_0, \ldots, u_{N-1}\}) - g(x_0) \end{cases}$$

in which g is the function $g(x_0) = -3x_0$.

Using the results of part (A), or otherwise, derive formulae for the leader's optimizing Stackelberg strategy and for the follower's response. [4]

Interpretation: The leader is a company whose manufacturing activity generates toxic waste x_0 at time $t = 0$, for a profit $g(x_0)$. The leader pays a follower, who uses a controlled industrial process over a time horizon $t = 0, \ldots, N$, to reduce the waste. The leader is charged for the minimum cost of reducing the waste. The leader then adjusts the waste level x_0, to maximize the leader's net profit $g(x_0) - \min_{\{u_0, \ldots, u_{N-1}\}}\{J(x_0, \{u_0, \ldots, u_{N-1}\})\}$.

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